

Instructor Pilot Checklist: Test Before Students Do

Almost every risk in this workshop is a "will it work on the lab computers?" question that cannot be answered from outside the lab. This checklist turns each one into a short test. Write the result next to each item.

Tested so far: only the things marked "Observed" below. Everything else is **not yet tested** on a Windows 11 Home computer.

Pilot A: Before Day 2 (installers and the terminal)

Use a Windows 11 Home PC that does **not** already have Node.js or Git. A lab PC is best.

A1. Node.js installs. Run the Node.js 22 LTS `.msi` from the share. Does it need an administrator prompt? Does it install without extra steps? Time taken: _____

A2. Git installs. Run the Git for Windows `.exe`. Which screens appear? Is the default editor Vim? Time taken: _____

A3. New terminal sees the tools. Close VS Code, open it again, open a **Command Prompt** terminal. Run `node -v`, `npm -v`, `git --version`. Results: _____

A4. Setup check. Copy `setup-check.js` into a folder and run `node setup-check.js`. Do you see four PASS lines? _____

A5. PowerShell behaviour. Open a **PowerShell** terminal and run `npm -v`. Do you get a "running scripts is disabled" message? _____ (If yes, the advice to use Command Prompt is confirmed. If no, it is still the safer default.)

A6. Windows prompts. Does Windows show "Windows protected your PC" for the installers? _____

A7. VS Code terminal profile. Can you pick **Command Prompt** from the terminal dropdown? _____

A8. LAN share speed. Copy the kit from the share to 5 computers at the same time. How long does it take? _____

A9. Lab policy. Do files and installed programs stay on the computers from day to day? _____
Where can students save backups? _____

Pilot B: Before Day 3 (Express and npm)

B1. npm install. In an empty folder run `npm init -y` then `npm install express`. Does it work on the lab network? Time taken: _____

B2. Many computers at once. Ask 5 to 10 computers to run `npm install express` at the same time. Does it slow down or fail? _____

B3. Offline plan. If B1 or B2 is slow, prepare a zipped `node_modules` folder (Express is plain JavaScript, so the same folder works on any computer) and put it on the share. Test that unzipping it into a project works: _____

Pilot C: Before Day 4 (SQLite)

C1. Built-in SQLite. On Node.js 22 LTS, run: `node -e "require('node:sqlite')"`. Does it work without any flag? _____ Does it print an "ExperimentalWarning"? _____

- **Observed so far:** on Node.js 22.22.0 on Linux, it worked without a flag and printed an ExperimentalWarning. The Node.js documentation says it is "no longer behind a flag but still experimental" from version 22.13. **Not yet tested on Windows.**

C2. Alternative driver. If C1 fails, test the `better-sqlite3` package: run `npm install better-sqlite3` on a lab PC and check that it installs without needing a compiler. _____ (Not tested.)

C3. Decide. Write the driver chosen for the workshop: _____

C4. PostgreSQL (optional track). Install the PostgreSQL installer on a lab PC. Does it install without problems? How long? _____ Does `npm install pg` work? _____ (Not tested.)

Pilot D: Before Day 5 (GitHub and Render)

D1. GitHub. Create a throwaway repository. In VS Code, run `git init`, `git add .`, `git commit`, and `git push`. Does it work from the lab network? _____

D2. Render sign-up. Sign up for Render with a new account. Does Render ask for a **credit card** or a **phone number** before it lets you create a Free web service? _____ (The documentation pages I read do not say.)

D3. Free web service from GitHub. Create a web service from your throwaway repository using the **Free** instance type. Write down the exact build command and start command you used, and how long the first deploy took: _____

D4. Node version on Render. Is the app built with Node.js 22? How did you set it? _____

D5. Environment variable. Add the `NOTES_PASSWORD` variable in the dashboard. Does the app read it? _____

D6. Cold start and data reset. Leave the service idle for more than 15 minutes, then open it. How long does the first load take? _____ Does a SQLite file created by the app disappear after a restart? _____ (Render's documentation says the filesystem is not kept.)

D7. Render PostgreSQL (optional track). Create a free PostgreSQL database. Note when it expires _____ (the documentation says 30 days after creation). Can the app connect with the connection string? _____ Is SSL needed? _____

D8. Many sign-ups. Ask 3 colleagues or students to sign up from the same network. Any extra checks or limits? _____

Pilot E: Before the first quiz (Google Forms and Classroom)

E1. Build the quiz. Follow `tools/quiz-builder/README.md` to build the Day 1 quiz from `00-Instructor-Kit/quiz/Day-1-Quiz.csv`. Does the script finish? ____ Is quiz mode on in the form? ____ Do answers and points show? ____

E2. Classroom. Create a quiz assignment in Google Classroom and attach the form. Write the exact menu path that worked: ____

E3. Take it. Take the quiz as a student account. Is the score shown? ____

If a test fails

If this fails	Fall back to
Installers cannot run on lab PCs	Pair students on computers where they can run
Node.js in the terminal is not found	Close VS Code, reopen it, open a new terminal; restart the computer
<code>npm install</code> is slow or fails	Use the zipped <code>node_modules</code> from the share
<code>node:sqlite</code> fails on Windows	Use <code>better-sqlite3</code> (Pilot C2)
Render asks for a card or phone number	Deploy only your own copy on the projector, and keep student apps running locally (Tier 1 and Tier 2 of the success tiers)
GitHub sign-up is slow from campus	Ask students to create accounts at home on Day 1 and Day 2
Google Forms script fails	Create the form by hand from the printable answer key